

ECON 3200 Discussion Note - Sep 9, 2026

Review: Gravity

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1 The gravity equation

The gravity equation used in lecture is

$$Trade_{AB} = T \frac{Y_A Y_B}{d_{AB}^\alpha} \varepsilon_{AB},$$

where:

- $Trade_{AB}$ is trade between countries A and B ;
- Y_A and Y_B are the two countries' GDPs;
- d_{AB} is the distance between the two countries;
- T is a constant;
- $\alpha > 0$ governs how strongly distance reduces trade;
- ε_{AB} captures other factors affecting bilateral trade.

The equation can also be written as

$$Trade_{AB} = T Y_A Y_B d_{AB}^{-\alpha} \varepsilon_{AB}.$$

Taking natural logs gives

$$\ln Trade_{AB} = \ln T + \ln Y_A + \ln Y_B - \alpha \ln d_{AB} + \ln \varepsilon_{AB}.$$

2 Interpreting the gravity coefficients

The gravity equation tells us how trade responds proportionally to GDP and distance.

GDP

The coefficient on $\ln Y_A$ is 1, and the coefficient on $\ln Y_B$ is also 1. Therefore,

$$\boxed{\text{GDP elasticity of trade} = 1.}$$

Holding everything else constant, a 1% increase in either country's GDP is associated with approximately a 1% increase in bilateral trade.

For example, if country B 's GDP rises by 10%, predicted trade between A and B rises by approximately 10%, holding distance and other factors fixed.

Distance

The coefficient on $\ln d_{AB}$ is

$$-\alpha.$$

Therefore,

$$\boxed{\text{distance elasticity of trade} = -\alpha.}$$

Lecture notes report estimates of approximately

$$\alpha \approx 1.$$

So the empirical gravity relationship implies approximately

$$\boxed{\text{a 1\% increase in distance is associated with a 1\% decrease in trade.}}$$

The word *associated* is important: the gravity equation describes a strong empirical relationship between distance and trade.

3 Approximate versus exact percentage changes

Suppose the distance elasticity of trade is -1 . For a small percentage change in distance,

$$\frac{\Delta \text{Trade}}{\text{Trade}} \approx -\frac{\Delta d}{d}.$$

So if distance increases by 10%, elasticity tells us that trade falls by approximately 10%.

However, the exact change is slightly different. If

$$d' = 1.1d,$$

then

$$Trade' = K(1.1d)^{-1},$$

where K collects all other terms. Therefore,

$$\frac{Trade'}{Trade} = \frac{K(1.1d)^{-1}}{Kd^{-1}} = \frac{1}{1.1} \approx 0.909.$$

Thus trade falls by about

$$1 - 0.909 = 0.091,$$

or 9.1%.

Key idea

Elasticity gives a convenient local approximation:

$$\boxed{\% \Delta y \approx \varepsilon_{y,x} \times \% \Delta x}$$

For small percentage changes, this approximation is usually very good.

4 Why do we need to control for country size?

Suppose we want to study the relationship between U.S. trade and distance.

Starting from the gravity equation,

$$\ln Trade_{US,B} = \ln T + \ln Y_{US} + \ln Y_B - \alpha \ln d_{US,B} + \ln \varepsilon_{US,B}.$$

For a given year, Y_{US} is the same for every U.S. trading partner, so

$$\ln T + \ln Y_{US}$$

is a constant.

But Y_B differs greatly across trading partners. Large countries tend to trade more simply because their economies are larger.

To isolate the relationship between trade and distance, move $\ln Y_B$ to the left-hand side:

$$\boxed{\ln Trade_{US,B} - \ln Y_B = \underbrace{\ln T + \ln Y_{US}}_{\text{constant}} - \alpha \ln d_{US,B} + \ln \varepsilon_{US,B}.}$$

This allows us to examine the relationship between trade and distance after accounting for the size of country B .

Economic intuition

If we simply compare raw trade with distance, differences in trade could reflect:

1. differences in distance;
2. differences in the size of the trading partner.

The gravity equation separates these two effects.

5 Economic interpretation of gravity

The gravity equation summarizes two central empirical patterns:

Larger economies trade more.

and

More distant economies trade less.

In compact form,

$$Trade_{AB} \propto Y_A Y_B d_{AB}^{-\alpha}.$$

So bilateral trade is larger when:

- country A is larger;
- country B is larger;
- the two countries are closer together.

This is why the model is called the *gravity model*: it resembles Newton's gravity equation, where attraction is stronger between larger objects and weaker when they are farther apart.

Summary

For the gravity equation

$$Trade_{AB} = T Y_A Y_B d_{AB}^{-\alpha} \varepsilon_{AB},$$

the key elasticity interpretations are

$$\boxed{\varepsilon_{Trade,Y_A} = 1, \quad \varepsilon_{Trade,Y_B} = 1, \quad \varepsilon_{Trade,d} = -\alpha.}$$

With

$$\alpha \approx 1,$$

a 1% increase in distance is associated with approximately a 1% decrease in trade.

When studying the relationship between trade and distance, we control for GDP because larger economies naturally trade more. After controlling for country size, the slope of the log-distance relationship identifies the distance elasticity of trade.